

Karla D - Class 05: Data Visualization with GGLOT

Q1 - For which phases is data visualization important in our scientific workflows? Answer- All of the above

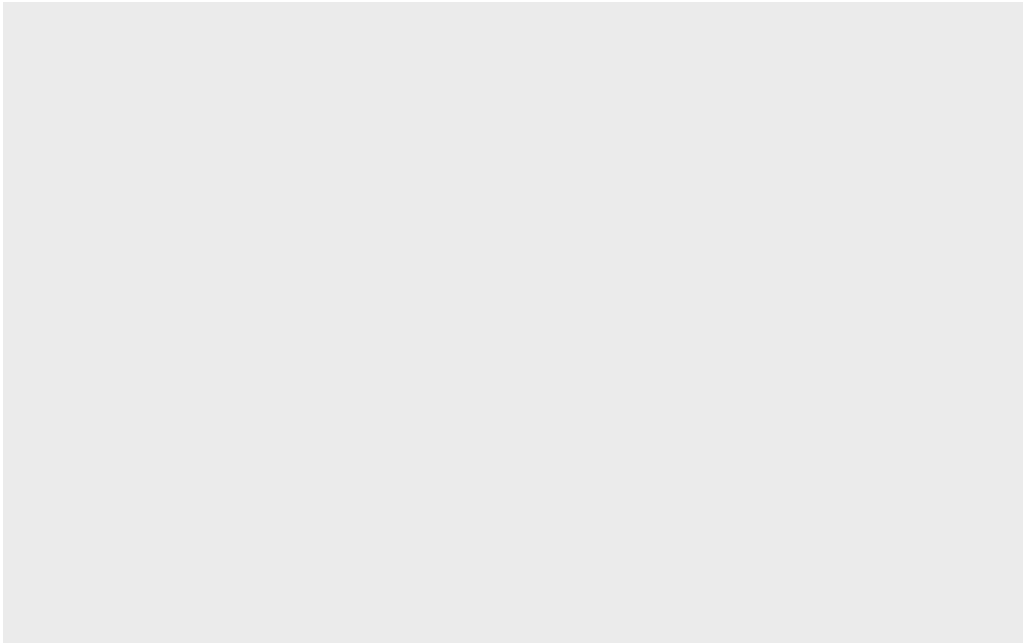
Q2- True or False? The ggplot2 package comes already installed with R? Answer- FALSE

Q3- Which plot types are typically NOT used to compare distributions of numeric variables? Answer - Network graphs

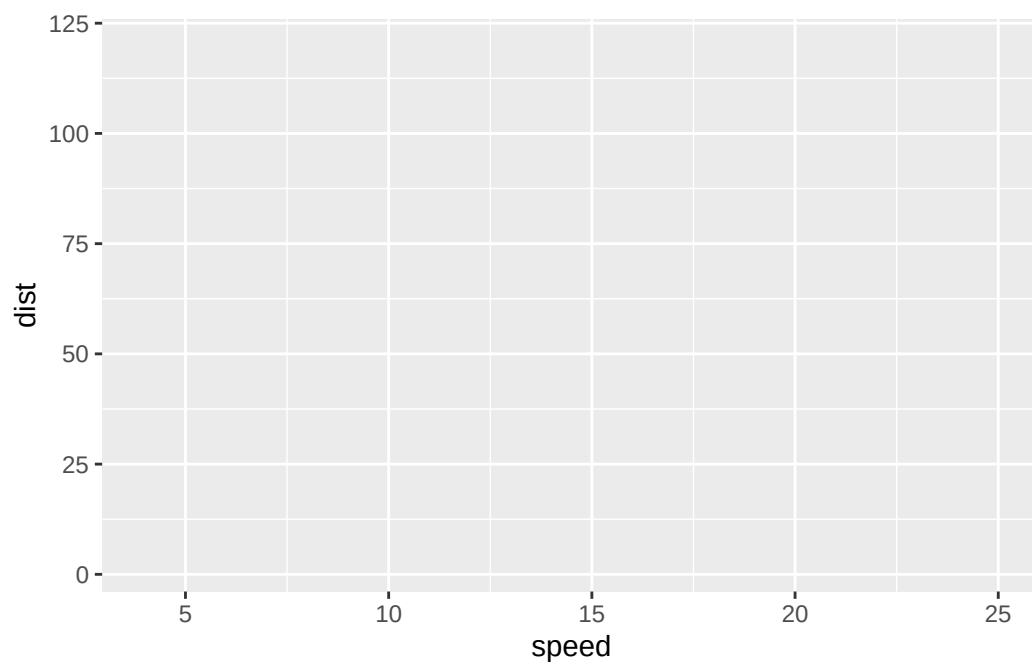
Q4- Which statement about data visualization with ggplot2 is incorrect? Answer- ggplot2 is the only way to create plots in R

```
library(ggplot2)
```

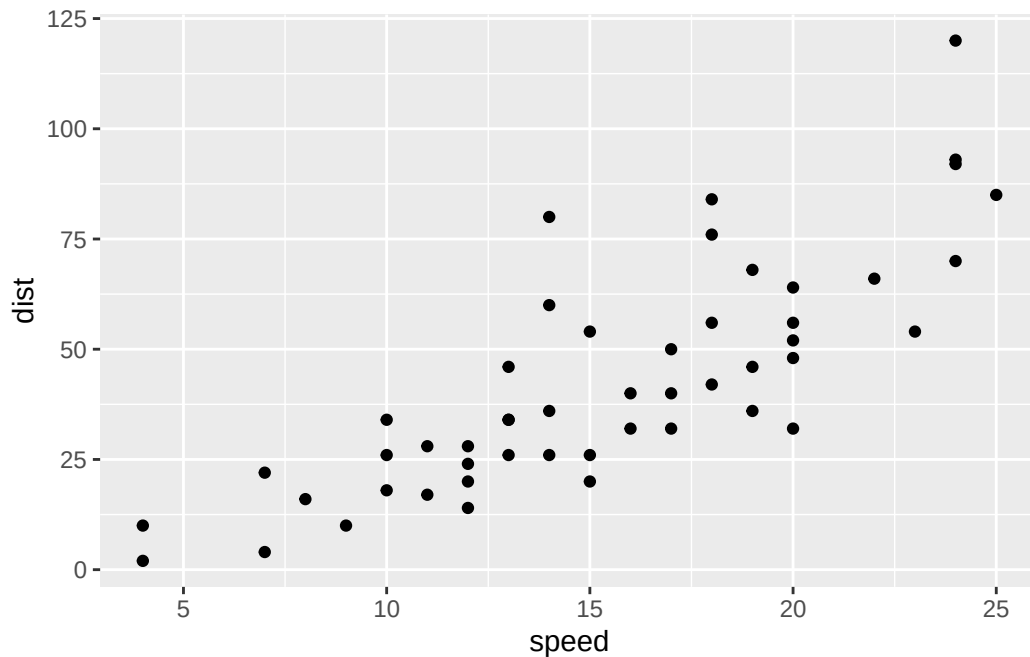
```
library(ggplot2)  
ggplot(cars)
```



```
ggplot(cars) +  
  aes(x=speed, y=dist)
```



```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()
```



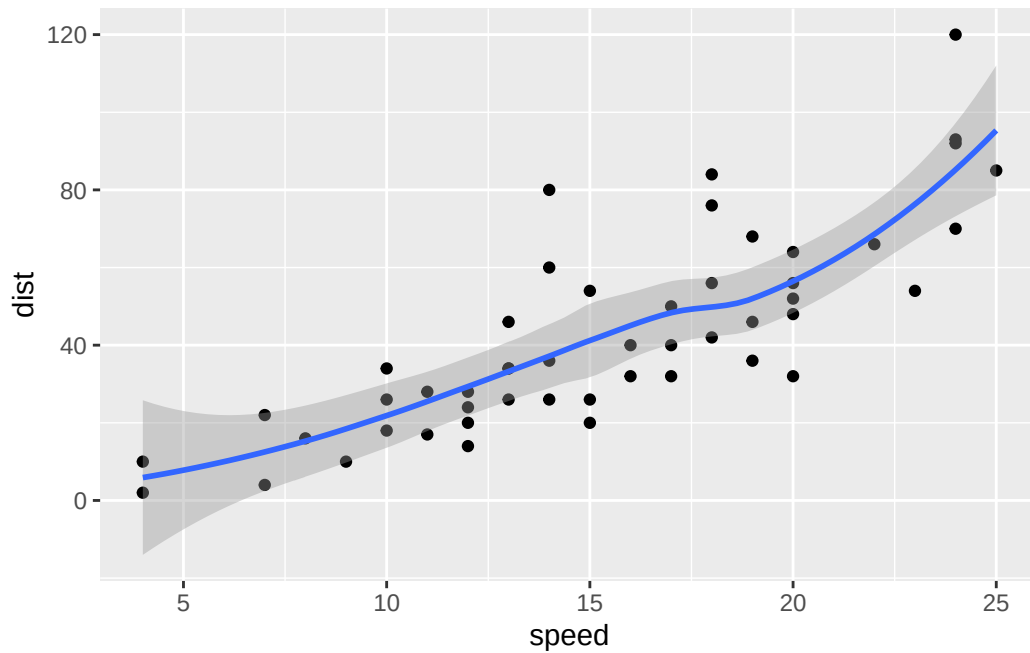
Q5 - Which geometric layer should be used to create scatter plots in ggplot2? Answer - `geom_point()`

Q6 - In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

Answer-

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth()
```

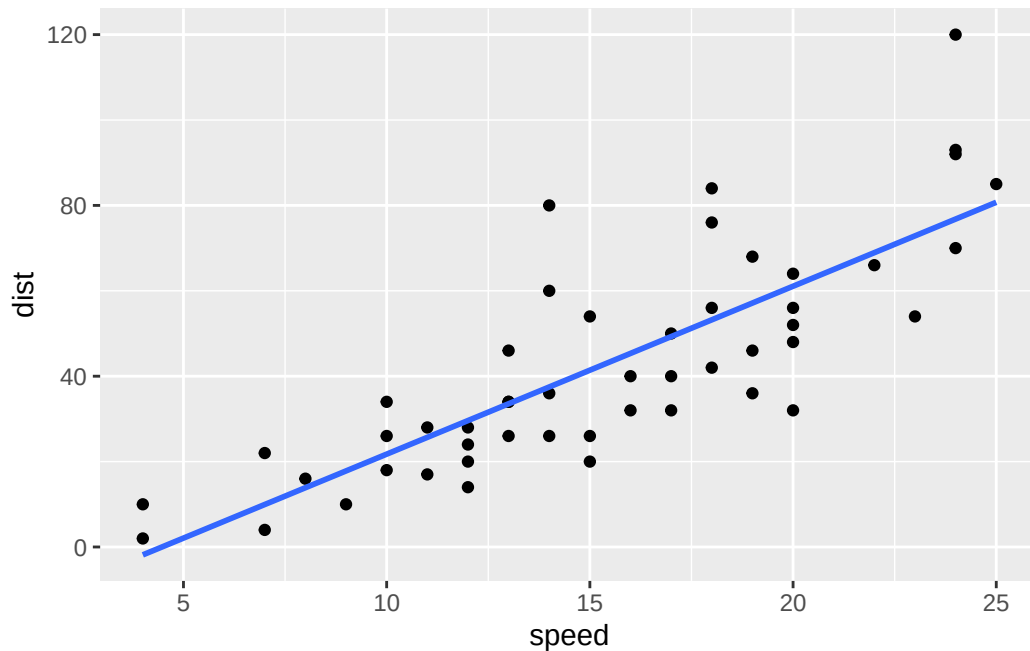
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



Q7- Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region? Answer-

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)
```

``geom_smooth()`` using formula = 'y ~ x'



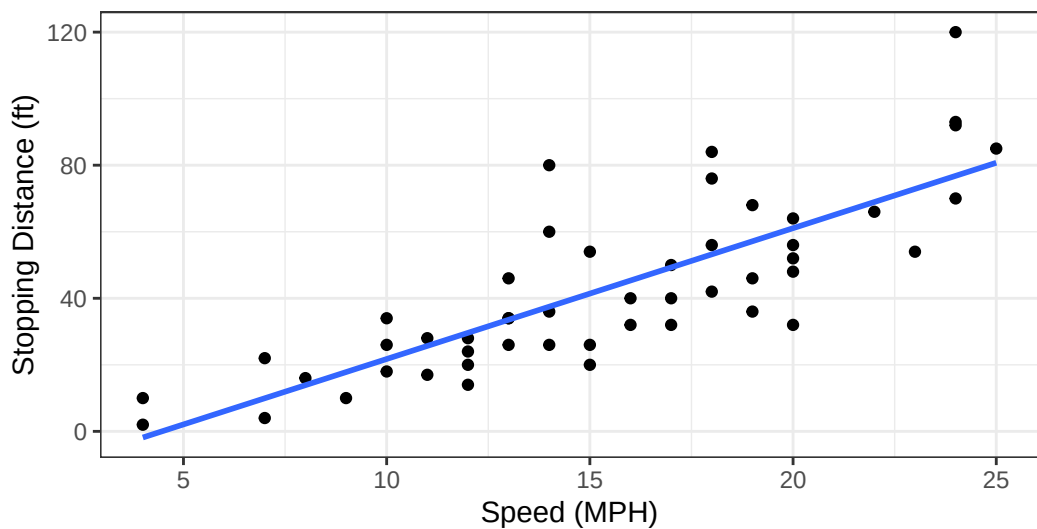
Q8- Can you finish this plot by adding various label annotations with the `labs()` function and changing the plot look to a more conservative “black & white” theme by adding the `theme_bw()` function: Answer-

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  labs(title="Speed and Stopping Distances of Cars",
       x="Speed (MPH)",
       y="Stopping Distance (ft)",
       subtitle = "Your informative subtitle text here",
       caption="Dataset: 'cars'") +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw()
```

``geom_smooth()`` using formula = 'y ~ x'

Speed and Stopping Distances of Cars

Your informative subtitle text here



Dataset: 'cars'

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
nrow(genes)
```

```
[1] 5196
```

```
colnames(genes)
```

```
[1] "Gene"      "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

```
table(genes$State)
```

down	unchanging	up
72	4997	127

```
round( table(genes$State)/nrow(genes) * 100, 2 )
```

down	unchanging	up
1.39	96.17	2.44

Q10- Use the nrow() function to find out how many genes are in this dataset. What is your answer? Answer- 5196

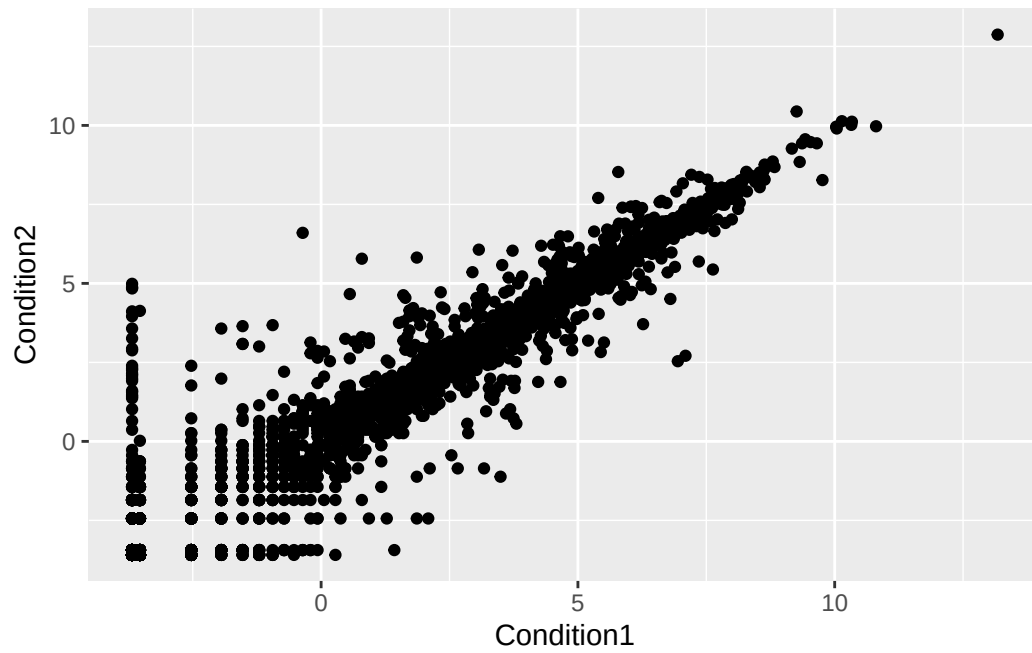
Q11- Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find? Answer- 4

Q12- Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer- 127

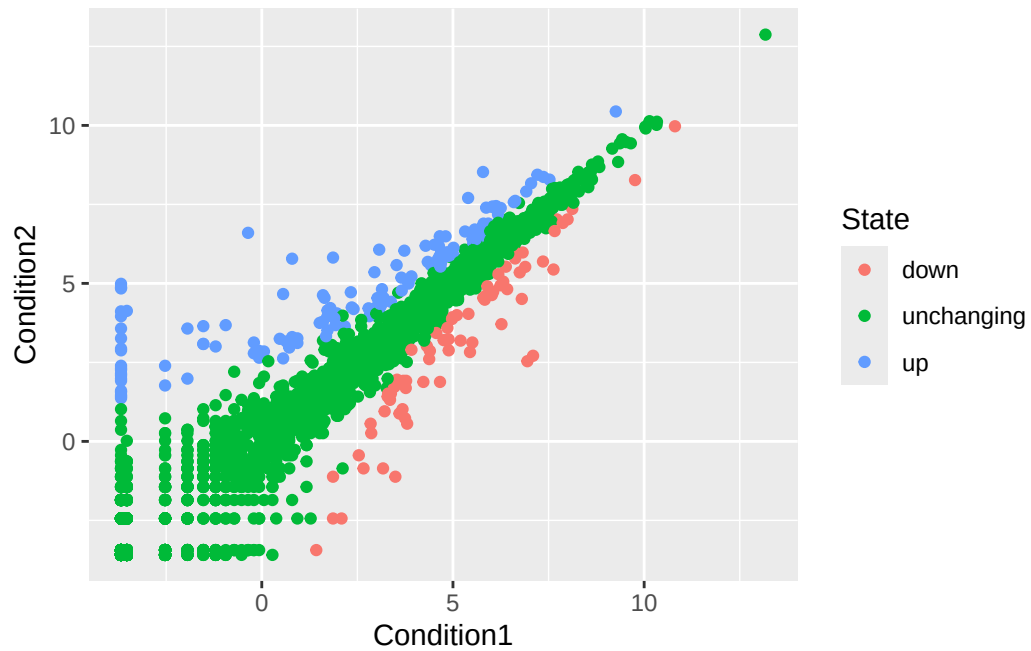
Q13- Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset? Answer- 2.44

Q13- Complete the code below to produce the following plot Answer-

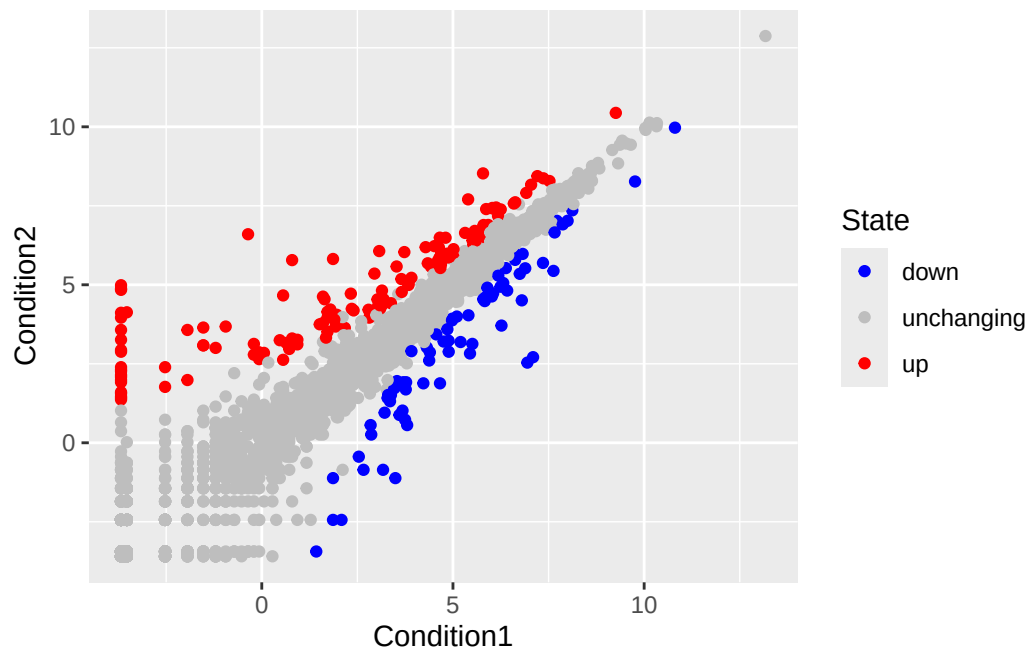
```
ggplot(genes) +  
  aes(x=Condition1, y=Condition2) +  
  geom_point()
```



```
p <- ggplot(genes) +  
  aes(x=Condition1, y=Condition2, col=State) +  
  geom_point()  
p
```

```
p + scale_colour_manual( values=c("blue","gray","red") )
```



Q13- Nice, now add some plot annotations to the p object with the labs() function so your plot looks like the following:

```
p + scale_colour_manual(values=c("blue","gray","red")) +
  labs(title="Gene Expression Changes Upon Drug Treatment",
        x="Control (no drug) ",
        y="Drug Treatment")
```

